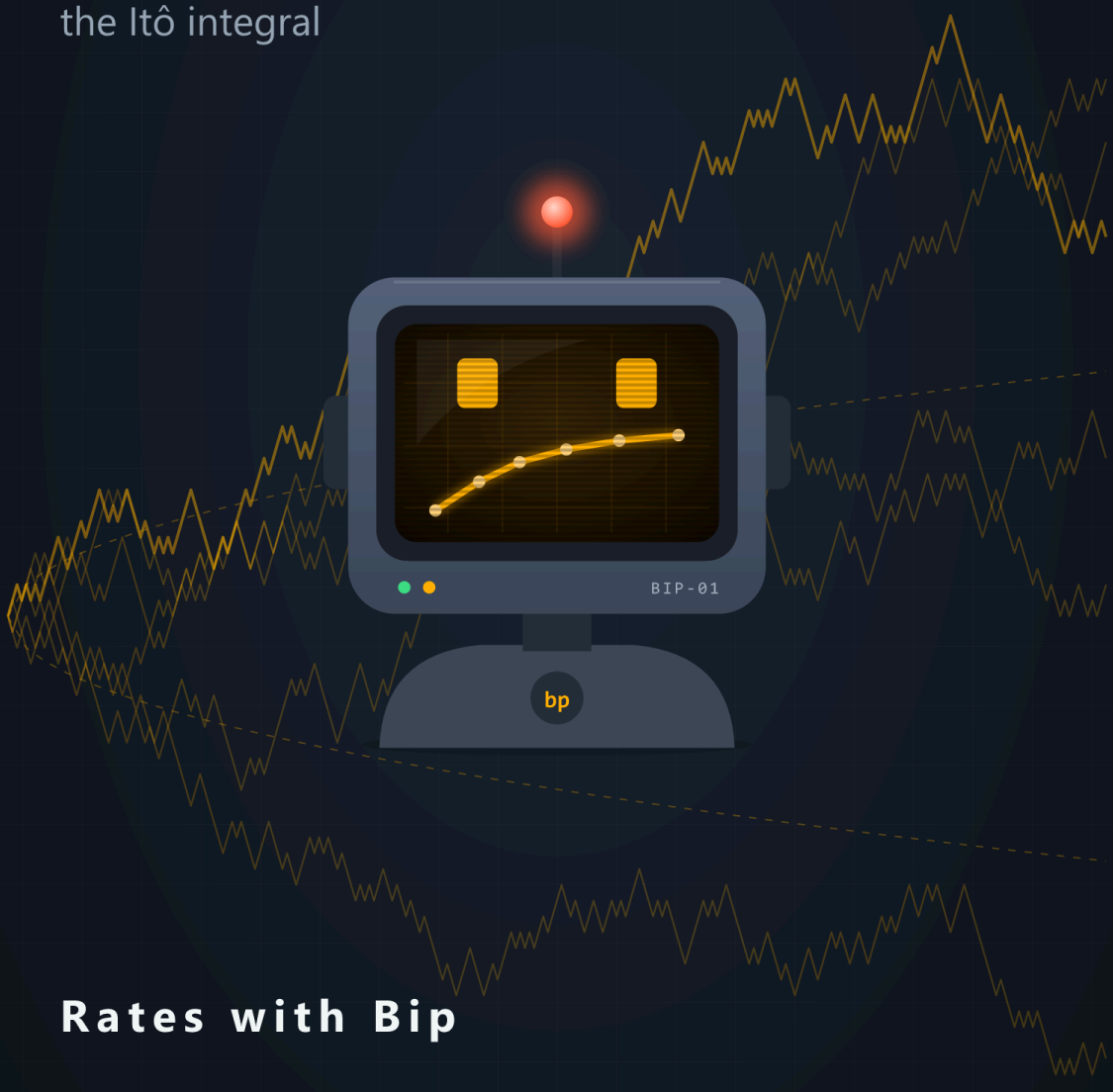
QUANT SERIES · PART 2 OF 4

Stochastic Calculus from Zero

Part 2: Brownian motion and the Itô integral



Rates with Bip

STOCHASTIC CALCULUS FROM ZERO · PART 2 OF 4

Stochastic Calculus from Zero

Part 2: Brownian motion and the Itô integral, with Python

Rates with Bip

Stochastic Calculus from Zero. Part 2: Brownian motion and the Itô integral, with Python.

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Every Brownian path in this book is simulated with numpy and a fixed seed. The share in Chapters 3 and 7 (volatility 20%, real-world drift 8%) and the correlation of 0.5 in Chapter 8 are illustrative, made up for teaching. There is no market data.

The history in Chapters 1, 2, 5 and 7 is checked against the original publications through their Crossref records (Paley, Wiener and Zygmund 1933; Itô 1944 and 1951; Stratonovich 1966), the Académie des sciences homage to Doeblin, and secondary sources for Donsker's theorem and Lévy's characterisation; sources are listed at the end of each chapter.

Every number in the text, tables and figures is computed by the companion Python notebooks (numpy only: Part 2 needs no QuantLib), one per chapter, which reproduce each result. Every chapter's exercises are in its notebook too, and the code solutions check themselves.

This book is for education only. It is not financial, investment, legal or tax advice.

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How to use this book

This book follows the eight video episodes of Part 2 of *Stochastic Calculus from Zero* (Episodes 9 to 16), one chapter per episode, and goes further in each: the definitions in full, and the short proofs the videos only sketch. It assumes Part 1: worlds, filtrations, conditional expectation, martingales and changes of odds, all on a coin tree. Part 2 carries those ideas over to continuous paths.

One puzzle runs through the whole series. Part 1's first chapter showed the chain rule from school failing on a random path, by exactly the sum of the squared steps. Part 2 builds the tools to fix it, and fixes it twice: for $\int W\,dW$ in Chapter 5, and in general, as the Itô-Doeblin formula, in Chapter 7. Both are checked on the very path that posed the puzzle.

- **Chapter 1** defines Brownian motion by four properties and shows why scaled random walks converge to it.
- **Chapter 2** shows its paths have no slope anywhere, look the same at every scale, and have infinite length.
- **Chapter 3** proves that squared steps add up to time, and draws the consequence: volatility can be measured, drift can't.
- **Chapter 4** builds martingales from Brownian motion, and meets the corrections Itô's formula explains.
- **Chapters 5 and 6** build the Itô integral and prove its two key properties: mean zero, and the Itô isometry.
- **Chapter 7** derives the Itô-Doeblin formula and checks it path by path.
- **Chapter 8** adds a second, correlated Brownian motion.

Each chapter ends with its one idea to remember (*Bip's rule*) and a question answered at the start of the next chapter. The back matter has all forty exercises with full answers, the Part 2 checkpoint, a glossary and further reading.

Paths you can reproduce

Every number, table and figure comes from a companion Python notebook, one per chapter, using numpy only. Every simulation uses the seed 2026, and the first two paths that seed draws reappear across chapters: **Episode 1's path** (1,000 steps, ending

at $W_1 = 0.666$) and a fine path of 2^{20} steps, called **Episode 10's path** here. Seeing the same paths again is deliberate: each new tool is tried on familiar ground.

The notebooks are free at github.com/RatesWithBip/curriculum, in the `stochastic-calculus-from-zero` folder, each with an *Open in Colab* button. Each ends with its chapter's exercises, whose code solutions check themselves.

Reading alongside

Steven Shreve's *Stochastic Calculus for Finance II*, Chapters 3 and 4, covers the same ground with full measure-theoretic proofs, and Bernt Øksendal's *Stochastic Differential Equations*, Chapters 2 to 4, gives a compact alternative. This book is written fresh from its own notebooks and uses none of their examples or exercises.

Free, and staying free

The videos, the notebooks and this book are free. If they help you, you can support the channel with a coffee at ko-fi.com/rateswithbip.

This book is for education only and is not financial advice.

CHAPTER 1 · VIDEO EPISODE 9

Brownian Motion, Defined

Four properties pin Brownian motion down: it starts at zero, its increments are independent and normal with variance equal to the elapsed time, and its paths are continuous. Scaled random walks converge to it.

Last chapter's question

The answer is $p = q$, here 0.55. If $Z = 1$ in every world, the real-world and risk-neutral weights agree: the share is expected to grow at exactly the deposit rate, and its risk earns no premium.

The definition

A **Brownian motion** (or Wiener process) is a family of random variables $(W_t)_{t \geq 0}$ such that

1. $W_0 = 0$;
2. for $0 \leq s < t$, the increment $W_t - W_s$ is independent of $\{W_u : u \leq s\}$ (**independent increments**);
3. $W_t - W_s$ is normal with mean 0 and variance $t - s$;
4. $t \mapsto W_t$ is continuous on every path.

Its filtration $\mathcal{F}_t$ is the information generated by the path up to time t, the continuous-time version of Part 1's blocks of worlds. On a computer we simulate W on a grid by adding independent normal steps of variance Δt, which gives properties 1 to 3 exactly at the grid points. Episode 1's path, the first seed-2026 draw, ends at $W_1 = 0.666$.

Why scaled random walks converge

Take a fair ± 1 walk M_n and scale it to $W_t^{(n)} = M_{\lfloor nt \rfloor} / \sqrt{n}$, so that its variance at time 1 is exactly 1. At $t = 1$ its distribution is binomial, and we can compare it exactly with the normal distribution function $N(x)$. The largest gap between the two (the **Kolmogorov distance**) shrinks like $1/\sqrt{n}$:

Exact binomial distribution function of the scaled walk at time 1 against the normal one.

STEPS OF THE WALK	LARGEST GAP TO N(X)
10	0.1230
100	0.0398
1,000	0.0126
10,000	0.0040

That is the central limit theorem at one time. **Donsker's theorem** (1951), also called the functional central limit theorem, says much more: the whole scaled path $W^{(n)}$ converges in distribution, as a random continuous function, to Brownian motion. This is the precise sense in which Brownian motion is "the limit of random walks".

Covariance

For $s \le t$,

$$\operatorname{Cov}(W_s, W_t) = \operatorname{Cov}\big(W_s,\ W_s + (W_t - W_s)\big)$$

$$= \operatorname{Var}(W_s) + \operatorname{Cov}(W_s,\ W_t - W_s) = s + 0,$$

because the increment is independent of W_s. So $\operatorname{Cov}(W_s, W_t) = \min(s, t)$.

20,000 simulated paths.

S	T	SIMULATED COVARIANCE	MIN(S, T)
0.25	0.75	0.248	0.25
0.50	0.50	0.498	0.50
0.50	1.00	0.495	0.50
0.75	1.00	0.744	0.75

The simulated correlation of $W_{0.25}$ with the later increment $W_{0.75} - W_{0.25}$ is -0.006, within the noise of zero. In fact Brownian motion can be defined equivalently as the continuous, mean-zero **Gaussian process** with covariance $\min(s, t)$: all its finite-dimensional distributions are jointly normal, and that covariance determines them.

Scaling

For any $c > 0$, the process $\tilde{W}_t = W_{ct}/\sqrt{c}$ is again a Brownian motion. Check the four properties: $\tilde{W}_0 = 0$; its increments are increments of W over disjoint intervals, so independent; $\tilde{W}_t - \tilde{W}_s = (W_{ct} - W_{cs})/\sqrt{c}$ is normal with variance $c(t-s)/c = t - s$; and

it is continuous. In simulation, $W_{0.25}/\sqrt{0.25}$ has variance 0.998. Chapter 2 shows what scaling implies for the shape of the paths.

Not just a walk with small steps

A random walk moves by exactly $\pm$ a fixed size; Brownian motion has normal increments over every interval, however short. No random walk *is* a Brownian motion; Donsker's theorem says they become one in the limit.

Bip's rule. Four properties pin Brownian motion down.

Try this. What is $\mathrm{Cov}(W_2, W_5)$?

Sources

Companion notebook: `s5e09_brownian_motion_defined.ipynb`. Every number in this chapter is computed by it.

1. Wikipedia, *Donsker's theorem (functional central limit theorem); M. D. Donsker (1951), An invariance principle for certain probability limit theorems, Memoirs of the AMS 6*. https://en.wikipedia.org/wiki/Donsker%27s_theorem
2. Wikipedia, *Wiener process: definition, Donsker's theorem (Wiener process as a limit of a random walk)*. https://en.wikipedia.org/wiki/Wiener_process

CHAPTER 2 · VIDEO EPISODE 10

Brownian Paths Are Strange

A Brownian path is continuous everywhere and differentiable nowhere. Its slope over a window of length h is about one over the square root of h, it looks the same at every scale, and its length is infinite.

Last chapter's question

The answer is 2: $\mathrm{Cov}(W_2, W_5) = \min(2, 5)$.

The slope that isn't

Take the slope of a path over a window of length h, $(W_{t+h} - W_t)/h$. The numerator is normal with standard deviation $\sqrt{h}$, so the slope has standard deviation $1/\sqrt{h}$, and its average size is

$$E\left|\frac{W_{t+h} - W_t}{h}\right| = \frac{1}{h}\sqrt{\frac{2h}{\pi}} = \sqrt{\frac{2}{\pi h}} \;\to\; \infty.$$

On Episode 10's path of 2^{20} steps:

Shrink the window and the slope grows instead of settling on a derivative.

WINDOW 1/H	AVERAGE SIZE OF THE SLOPE	√(2/(πH))
8	2.3	2.3
128	9.3	9.0
1,024	25.6	25.5
16,384	101.4	102.1

A smooth function's difference quotients converge to its derivative. These blow up. In 1933 Paley, Wiener and Zygmund proved the strong statement: with probability one, a Brownian path has a derivative at **no** time at all.

Zoom in

Scaling (Chapter 1) says that the first fraction c of the path, stretched by $1/c$ in time and by $1/\sqrt{c}$ in height, is again a Brownian motion on $[0, 1]$.



Figure 2.1. Episode 10's path: the whole year, its first hundredth, and its first ten-thousandth, each stretched to fill the frame.

Zoom in on a smooth curve and it flattens into a straight line, its tangent. A Brownian path never does: it is **self-similar**. Over 20,000 paths, the first hundredth stretched ten times in height has variance 0.989 at its end, the same as a whole year.

Even its zeros cluster. On this grid the path changes sign 1,093 times in the year, 27 times in its first hundredth and 23 times in its first ten-thousandth. A Brownian path started at 0 returns to 0 infinitely often in every interval $(0, \varepsilon)$, and finer grids find ever more crossings.

Infinite length

The length of the path measured with n steps is $\sum |\Delta W_i|$, with mean $n\sqrt{2/(\pi n)} = \sqrt{2n/\pi}$:

Episode 10's path, measured on coarser and finer grids.

STEPS	LENGTH OF THE PATH	√(2N/Π)
10	3.2	2.5
1,000	24.9	25.2
1,048,576	816.9	817.0

A short proof that the length can't be finite uses Chapter 3. For any path,

$$\sum (\Delta W_i)^2 \;\le\; \max_i |\Delta W_i| \cdot \sum |\Delta W_i|.$$

If the length $\sum |\Delta W_i|$ stayed bounded, the right side would go to 0 as the grid shrinks, because a continuous path on a closed interval has $\max_i |\Delta W_i| \to 0$. But the left side tends to $T > 0$ (Chapter 3). So the length is infinite.

Why it matters

The integral of ordinary calculus, $\int f\, dg$ (the Riemann-Stieltjes integral), is built for integrators g of finite length. Along a Brownian path that construction fails: different ways of sampling f give different limits. Chapter 5 builds a new integral that works, and the choice it makes is the heart of stochastic calculus.

Roughness is not a flaw in the model, either. It is exactly what makes the squared steps add up to time, and that is what lets you measure volatility from one path (Chapter 3).

Bip's rule. Continuous everywhere, smooth nowhere.

Try this. Halve the window h 4 times. By how much does the typical slope grow?

Sources

Companion notebook: `s5e10_strange_paths.ipynb`. Every number in this chapter is computed by it.

1. Springer (Crossref record), *Notes on random functions (R. E. A. C. Paley, N. Wiener, A. Zygmund), Mathematische Zeitschrift 37 (1933), 647-668*. https://doi.org/10.1007/BF01474606
2. Wikipedia, *Wiener process: continuous everywhere but nowhere differentiable*. https://en.wikipedia.org/wiki/Wiener_process

CHAPTER 3 · VIDEO EPISODE 11

Quadratic Variation: What You Can Measure

Squared steps add up to the elapsed time, while products with a time step vanish. That multiplication table drives stochastic calculus, and it explains why a year of prices pins down volatility but hardly tells you the drift.

Last chapter's question

The answer is ×4. Halving h four times divides it by 16, and the slope grows like $1/\sqrt{h}$.

Quadratic variation

For a partition $0 = t_0 < t_1 < \cdots < t_n = T$ with mesh $\max \Delta t_i \to 0$, the **quadratic variation** of W on $[0, T]$ is the limit of $\sum (\Delta W_i)^2$. It equals T:

$$[W]_T = \lim \sum (W_{t_{i+1}} - W_{t_i})^2 = T.$$

Proof (in mean square). Let $Q_n = \sum (\Delta W_i)^2$. Each $(\Delta W_i)^2$ has mean Δt_i and variance $2(\Delta t_i)^2$, and they are independent, so

$$E[Q_n] = \sum \Delta t_i = T,$$

$$E[(Q_n - T)^2] = \operatorname{Var}(Q_n) = 2 \sum (\Delta t_i)^2 \le 2T \max_i \Delta t_i \to 0.$$

So $Q_n \to T$ in mean square. On Episode 10's path:

The first sum settles on the elapsed time; the other two vanish.

STEPS	Σ (ΔW)²	Σ |ΔW·ΔT|	Σ (ΔT)²
16	1.3708	0.2382	0.062500
256	0.9336	0.0487	0.003906
4,096	1.0035	0.0125	0.000244
1,048,576	1.0003	0.0008	0.000001

The other two sums vanish: $\sum |\Delta W_i|\,\Delta t_i \le \max|\Delta W_i| \cdot T \to 0$, and $\sum (\Delta t_i)^2 \le \max \Delta t_i \cdot T \to 0$. Written compactly, the **multiplication table**:

$$dW \cdot dW = dt, \qquad dW \cdot dt = 0, \qquad dt \cdot dt = 0.$$

Scaled paths and drift

For $X_t = \mu t + \sigma W_t$, each squared step is $\sigma^2(\Delta W)^2 + 2\mu\sigma\,\Delta t\,\Delta W + \mu^2(\Delta t)^2$, so by the table $[X]_T = \sigma^2 T$. The drift contributes nothing. On Episode 10's path with $\mu = 8\%$ and $\sigma = 20\%$ the sum is 0.0400 against $\sigma^2 = 0.04$.

Volatility you can measure

An illustrative share with $\sigma = 20\%$ and $\mu = 8\%$ has daily log returns $r_i = (\mu - \tfrac{1}{2}\sigma^2)\Delta t + \sigma\,\Delta W_i$ (Chapter 7 explains the $\tfrac{1}{2}\sigma^2$). Their sum of squares over a year approximates the quadratic variation of the log price, $\sigma^2 \times 1$, so the **realised volatility** $\hat{\sigma} = \sqrt{\sum r_i^2}$ estimates σ.

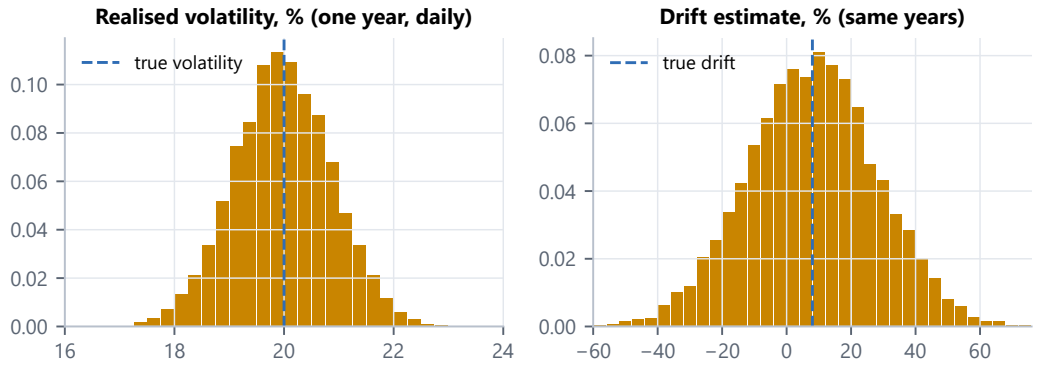


Figure 3.1. 10,000 simulated years of daily prices. Left: realised volatility. Right: the drift estimated from the same years.

Realised volatility averages 20.0% with standard deviation 0.9%, close to the theoretical $\sigma/\sqrt{2N}$ for $N = 252$ observations: each squared return has relative variance 2, and there are N of them.

Drift you can't

The drift estimate from the same year averages 8.0% but with standard deviation 20.0%. The reason is structural. The sum of the returns telescopes to the total log return over the year, $\ln(S_T/S_0) = (\mu - \tfrac{1}{2}\sigma^2)T + \sigma W_T$, so the only information about μ is

one normal draw with standard deviation $\sigma\sqrt{T}$. Dividing by T gives an error of $\sigma/\sqrt{T}$: 6.4% even after ten years.

Sampling more often

One year of data, 2,000 simulated years per row.

SAMPLING	OBSERVATIONS A YEAR	VOLATILITY ERROR	DRIFT ERROR
monthly	12	4.0%	20.5%
weekly	52	2.0%	19.8%
daily	252	0.9%	19.7%
every 15 minutes	6,552	0.2%	19.7%

Volatility lives in the squared increments, and finer sampling gives more of them; drift lives in the calendar. This is why the Black-Scholes price, which needs σ but not μ (*Options from Zero*, Part 2, Chapter 4), can be used at all. In practice volatility moves over time and returns have jumps, so real estimates are less clean than this, but the asymmetry remains.

Bip's rule. Volatility you can measure; drift you have to guess.

Try this. What is the quadratic variation of $3W$ over 2 years?

Companion notebook: `s5e11_quadratic_variation.ipynb`. Every number in this chapter is computed by it.

CHAPTER 4 · VIDEO EPISODE 12

Martingales from Brownian Motion

W is a martingale, W squared minus t is a martingale, and so is the exponential of sigma W minus half sigma squared t. The subtractions are the corrections that Itô's formula will explain.

Last chapter's question

The answer is 18: $3^2 \times 2$.

Three martingales

A process M is a martingale with respect to $(\mathcal{F}_t)$ if $E|M_t| < \infty$ and $E[M_t \mid \mathcal{F}_s] = M_s$ for $s \le t$: Part 1's definition, with continuous time. For $s < t$ write $W_t = W_s + D$ where $D = W_t - W_s$ is independent of $\mathcal{F}_s$, normal, mean 0, variance $t - s$. Using Part 1's rules (take out what is known; independence):

$$E[W_t \mid \mathcal{F}_s] = W_s + E[D] = W_s,$$

$$E[W_t^2 \mid \mathcal{F}_s] = W_s^2 + 2W_s E[D] + E[D^2] = W_s^2 + (t - s),$$

$$E[e^{\sigma W_t} \mid \mathcal{F}_s] = e^{\sigma W_s}\, E[e^{\sigma D}] = e^{\sigma W_s}\, e^{\frac{1}{2}\sigma^2 (t-s)}.$$

The last line uses the normal moment generating function, $E[e^{aZ}] = e^{\frac{1}{2}a^2 v}$ for $Z \sim N(0, v)$. Rearranging the second and third:

$$W_t^2 - t \quad \text{and} \quad \exp\!\left(\sigma W_t - \tfrac{1}{2}\sigma^2 t\right) \quad \text{are martingales.}$$

Averaged over 100,000 simulated paths with $\sigma = 20\%$:

Each stays at its starting value (0, 0 and 1) within the noise.

TIME	W	W² − T	EXP(ΣW − ½Σ²T)
0.5	-0.0006	0.0017	0.9999
1.0	-0.0005	0.0002	0.9999
2.0	0.0027	-0.0023	1.0005

Given the present

Constant averages are not enough (Part 1, Chapter 6): the best guess must hold *given what is known*. Group the paths by the value of $W_{0.5}$ and compare, within each group, the average of $W_1^2 - 1$ with the average of $W_{0.5}^2 - 0.5$:

In every group the later average matches today's value within three standard errors.

W AT HALF A YEAR	W² − T NOW	W² − T AT ONE YEAR	STANDARD ERROR	PATHS
-2.0 to -1.5	2.338	2.409	0.068	1,482
-1.5 to -1.0	0.962	0.941	0.023	6,268
-1.0 to -0.5	0.039	0.038	0.010	16,159
-0.5 to +0.0	-0.422	-0.418	0.005	25,752
+0.0 to +0.5	-0.422	-0.420	0.005	26,035
+0.5 to +1.0	0.035	0.038	0.010	16,236
+1.0 to +1.5	0.963	0.907	0.023	6,161
+1.5 to +2.0	2.345	2.390	0.068	1,459

Convexity and the exponential

Without the correction, $E[e^{\sigma W_t}] = e^{\frac{1}{2}\sigma^2 t}$ grows: 1.0201 at one year and 1.0413 at two, against $e^{0.02}$ and $e^{0.04}$. This is **Jensen's inequality**: for a convex f, $E[f(X)] \geq f(E[X])$, so $E[e^{\sigma W}] > e^0 = 1$. Subtracting $\frac{1}{2}\sigma^2 t$ in the exponent removes exactly the growth. The exponential martingale is the shape of a share price discounted under the risk-neutral odds, and of Girsanov's reweighting factor (Part 1, Chapter 8; Part 3).

The pattern

A straight-line function of a martingale is a martingale; a curved one picks up a drift. For W^2 the drift is 1 per unit time, for $e^{\sigma W}$ it is $\frac{1}{2}\sigma^2 e^{\sigma W}$: in both cases **half the second derivative**. Chapter 7 proves this is always so.

Bip's rule. Subtract the drift that randomness adds, and you have a martingale.

Try this. What do you subtract from W_t^3 to get a martingale?

Companion notebook: `s5e12_brownian_martingales.ipynb`. Every number in this chapter is computed by it.

CHAPTER 5 · VIDEO EPISODE 13

Building the Itô Integral

Along a Brownian path, where you sample the integrand changes the answer. Sampling at the left end of each step, as a trader must, gives the Itô integral, and for the integral of W dW it gives one half W squared minus one half T.

Last chapter's question

The answer is $3\int_0^t W_u\,du$. Since $E[W_t^3 \mid \mathcal{F}_s] = W_s^3 + 3W_s(t-s)$, the cube drifts at $3W$ per unit time; Chapter 7 gets this in one line.

Three ways to sample

Chapter 2 showed that the integral of ordinary calculus does not work along a Brownian path. Build one from sums instead: for a partition of $[0, T]$, add up $W_\star \Delta W_i$, where $W_\star$ is the path's value at the **left** end of the step, the **middle** (the average of the two ends) or the **right** end.

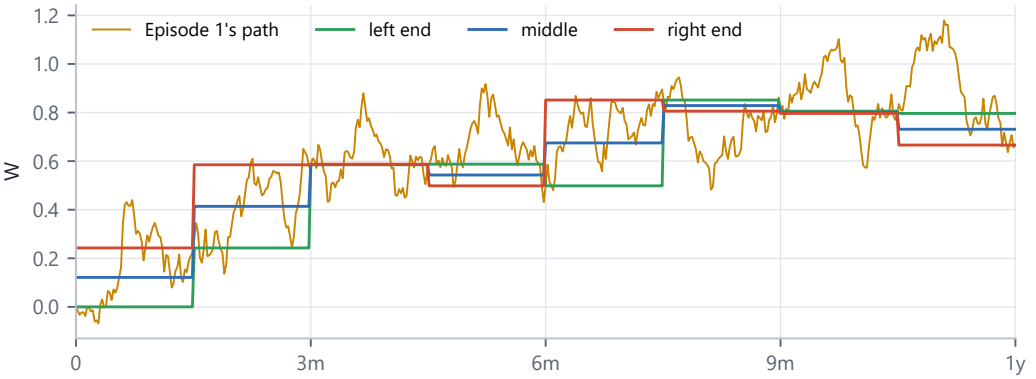


Figure 5.1. Episode 1's path, cut into 8 steps for the picture, with the three sampling rules drawn as step functions.

With all 1,000 steps of Episode 1's path ($W_1 = 0.666$):

Three rules, three answers, on the same path.

SAMPLE W AT	Σ W × ΔW
Left end	-0.3092
Middle	0.2219
Right end	0.7531

Three identities

These hold exactly, on any path, for any partition:

1. **Middle.** $\frac{1}{2}(W_i + W_{i+1})(W_{i+1} - W_i) = \frac{1}{2}(W_{i+1}^2 - W_i^2)$, so the sum telescopes to $\frac{1}{2}W_T^2$ (0.2219 here).
2. **Right minus left.** $(W_{i+1} - W_i)\,\Delta W_i = (\Delta W_i)^2$, so right − left $= \sum(\Delta W_i)^2$ (1.0623 here).
3. **Left.** Combining, left $= \frac{1}{2}W_T^2 - \frac{1}{2}\sum(\Delta W_i)^2$ and right $= \frac{1}{2}W_T^2 + \frac{1}{2}\sum(\Delta W_i)^2$.

Part 1's opening puzzle is now transparent: Episode 1's chain-rule sum, -0.6185, was exactly twice the left sum, and its gap from W_T^2 was the sum of squared steps.

The Itô integral of W dW

By Chapter 3, $\sum(\Delta W_i)^2 \to T$ in mean square, so the left sums converge:

$$\int_0^T W\,dW = \tfrac{1}{2}W_T^2 - \tfrac{1}{2}T \qquad \text{(Itô)}.$$

Episode 10's path, measured on coarser and finer grids.

STEPS	LEFT SUM − (½W² − ½T)
16	-0.1854
256	0.0332
4,096	-0.0017
1,048,576	-0.0001

The middle sums converge to $\frac{1}{2}W_T^2$: the **Stratonovich** integral, written $\int W \circ dW$.

The general construction

A **simple** integrand is a process that is constant on each step, $H_t = H_i$ for $t_i \le t < t_{i+1}$, with each H_i known at time t_i ($\mathcal{F}_{t_i}$-measurable). Its integral is defined as

$$\int_0^T H\,dW = \sum_i H_i\,(W_{t_{i+1}} - W_{t_i}).$$

For a general **adapted** integrand H with $E\int_0^T H_t^2\,dt < \infty$, choose simple integrands $H^{(n)} \to H$ in mean square; their integrals then form a Cauchy sequence in mean square (by the isometry of Chapter 6) and converge to a limit, the **Itô integral** $\int_0^T H\,dW$. The left-point rule is built into the definition: H_i must be known at the *start* of its step.

Itô or Stratonovich?

Averaged over many paths, the two integrals behave very differently:

The Itô sum averages zero; the Stratonovich sum averages ½T.

SAMPLED AT	AVERAGE OVER 20,000 PATHS
Left end (Itô)	-0.005
Middle (Stratonovich)	0.495

Stratonovich keeps the chain rule from school ($\int W \circ dW = \frac{1}{2}W^2$), but its integrals are not fair games. Itô gives up the school chain rule and keeps the martingale property, which is exactly what pricing needs. Finance therefore uses Itô; physics, where the chain rule matters more, often uses Stratonovich. Itô introduced his integral in 1944; Stratonovich's version dates from 1966.

Bip's rule. Decide, then the market moves: that's the Itô integral.

Try this. What does the right-end sum converge to?

Sources

Companion notebook: `s5e13_ito_integral.ipynb`. Every number in this chapter is computed by it.

1. Project Euclid (Crossref record), *Stochastic integral (K. Itô), Proceedings of the Imperial Academy, Tokyo 20(8) (1944), 519-524*. https://doi.org/10.3792/pia/1195572786
2. SIAM (Crossref record), *A new representation for stochastic integrals and equations (R. L. Stratonovich), SIAM Journal on Control 4 (1966), 362-371*. https://doi.org/10.1137/0304028

CHAPTER 6 · VIDEO EPISODE 14

No Peeking, No Free Lunch

An Itô integral has mean zero and is a martingale, and its variance is the expected integral of the integrand squared. Read as trading, every strategy that uses only the past is a fair game.

Last chapter's question

The answer is $\frac{1}{2}W_T^2 + \frac{1}{2}T$: the right sum is $\frac{1}{2}W_T^2 + \frac{1}{2}\sum(\Delta W)^2$, and the squared steps tend to T.

Mean zero, and the martingale property

Take a simple adapted integrand H (Chapter 5). Each term of its integral has mean zero, by Part 1's rule of taking out what is known:

$$E[H_i\,\Delta W_i] = E\big[H_i\,E[\Delta W_i \mid \mathcal{F}_{t_i}]\big] = E[H_i \cdot 0] = 0.$$

The same computation, conditioned on $\mathcal{F}_s$ instead of taken unconditionally, shows that $I_t = \int_0^t H\,dW$ satisfies $E[I_t \mid \mathcal{F}_s] = I_s$: **the Itô integral is a martingale**. Both properties pass to general adapted integrands in the limit.

The Itô isometry

Square the sum: $\big(\sum H_i\Delta W_i\big)^2 = \sum_i H_i^2(\Delta W_i)^2 + 2\sum_{i<j} H_i\Delta W_i\,H_j\Delta W_j$. In each cross term, $H_i\Delta W_i H_j$ is known at time t_j and ΔW_j is independent of it with mean zero, so the cross terms have expectation zero. In each diagonal term, $E[H_i^2(\Delta W_i)^2] = E\big[H_i^2\,E[(\Delta W_i)^2 \mid \mathcal{F}_{t_i}]\big] = E[H_i^2]\,\Delta t_i$. Therefore

$$E\Big[\Big(\int_0^T H\,dW\Big)^2\Big] = E\Big[\int_0^T H_t^2\,dt\Big].$$

This is the **Itô isometry**: it says the map $H \mapsto \int H\,dW$ preserves mean-square size, which is what makes the limit construction of Chapter 5 work. For $H = W$, $E\int_0^1 W_t^2\,dt = \int_0^1 t\,dt = \frac{1}{2}$; 20,000 simulated paths give 0.479 (standard error 0.012).

Trading gains

Read H as a position in an asset whose price moves like W; then $\int H\,dW$ is the trading gain. Every strategy that decides its position from the past has gains with mean zero:

20,000 simulated paths of 500 steps over a year.

STRATEGY	AVERAGE GAIN
Hold W (the integral of W dW)	-0.003
Long if W > 0, short if W < 0	0.004

The second strategy, long one unit while $W > 0$ and short one while $W < 0$, has $H^2 = 1$, so by the isometry its gains have variance $\int_0^1 1\,dt = 1$; the simulation gives 0.990.

Peeking

Break the rule: choose $H_i = \text{sign}(\Delta W_i)$ after seeing the step. Then every step is a gain of $|\Delta W_i|$, and over N steps the gains average $N\sqrt{2\Delta t/\pi} = \sqrt{2NT/\pi}$: 17.85 over 500 steps (theory 17.84). Trading more often makes it grow without limit, because the path has infinite length (Chapter 2). The left-point rule of the Itô integral is exactly what rules this out.

Against pure Brownian noise, no timing rule built on the past makes money on average. To win you need an edge: a drift in your favour, or information others don't have. Mathematically, that is a dt term, or an integrand that isn't adapted.

Bip's rule. No peeking, no free lunch.

Try this. What is the variance of $\int_0^4 2\,dW$?

Companion notebook: `s5e14_ito_isometry.ipynb`. Every number in this chapter is computed by it.

CHAPTER 7 · VIDEO EPISODE 15

The Itô-Doeblin Formula

Expand a function of Brownian motion to second order, replace each squared step by a time step, and the chain rule gains one term, half the second derivative times dt. Part 1's opening puzzle, solved in general.

Last chapter's question

The answer is 16: by the isometry, $\int_0^4 2^2\,dt$.

The formula

Itô-Doeblin formula. For f with continuous second derivative,

$$f(W_T) - f(W_0) = \int_0^T f'(W_t)\,dW_t + \tfrac{1}{2}\int_0^T f''(W_t)\,dt,$$

or, in shorthand,

$$df(W) = f'(W)\,dW + \tfrac{1}{2}f''(W)\,dt.$$

Derivation. Over a partition of $[0, T]$, Taylor's theorem gives

$$f(W_{t_{i+1}}) - f(W_{t_i}) = f'(W_{t_i})\,\Delta W_i + \tfrac{1}{2}f''(W_{t_i})\,(\Delta W_i)^2 + R_i,$$

with a remainder $|R_i| \le C\,|\Delta W_i|^3$ on the bounded range of the path (for this sketch, assume f has a continuous third derivative; two are enough with a finer argument). Sum over i:

- $\sum f'(W_{t_i})\,\Delta W_i$ are left sums: they converge to the Itô integral $\int f'(W)\,dW$ (Chapter 5).
- $\sum \tfrac{1}{2}f''(W_{t_i})(\Delta W_i)^2$ converges to $\tfrac{1}{2}\int f''(W)\,dt$, because the squared steps add up like time (Chapter 3): replace $(\Delta W_i)^2$ by Δt_i, and the difference has mean zero and vanishing variance.
- $\sum |R_i| \le C \max_i |\Delta W_i| \sum (\Delta W_i)^2 \to 0 \times T = 0$.

The left side telescopes to $f(W_T) - f(W_0)$.

With time and drift

An **Itô process** is $dX = \mu_t\,dt + \sigma_t\,dW$ (adapted μ, σ). By the multiplication table, $(dX)^2 = \sigma_t^2\,dt$. For $f(t, x)$, the same expansion with one more variable gives

$$df(t, X_t) = f_t\,dt + f_x\,dX_t + \tfrac{1}{2} f_{xx}\,\sigma_t^2\,dt.$$

Checked on a path

For $f(x) = e^x$ on Episode 10's path, compare the exact change $e^{W_1} - e^{W_0}$ with the school chain rule $\sum e^W \Delta W$ and with Itô's $\sum e^W \Delta W + \frac{1}{2}\sum e^W \Delta t$:

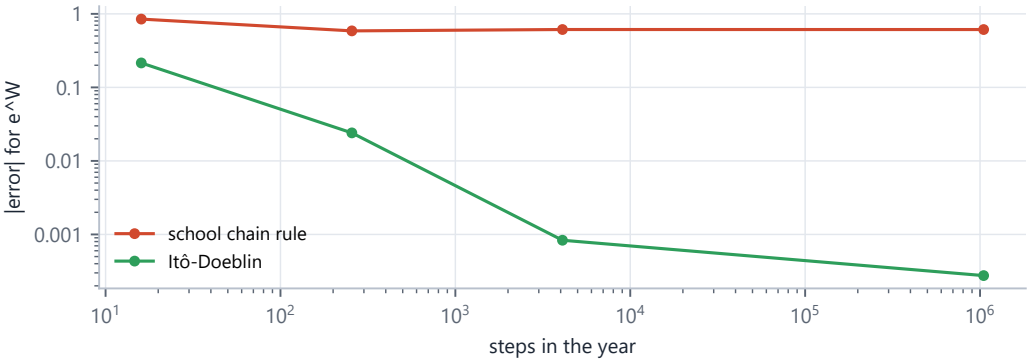


Figure 7.1. The chain rule's error settles at 0.61 and never shrinks; Itô's falls to 0.0003 on the finest grid.

Error = exact change minus the rule's prediction.

STEPS	CHAIN RULE: ERROR	ITÔ: ERROR
16	0.8490	0.2150
256	0.5868	-0.0241
4,096	0.6125	0.0008
1,048,576	0.6119	0.0003

Part 1's puzzle, solved

For $f(x) = x^2$: $f' = 2x$, $f'' = 2$, so

$$W_T^2 = 2\int_0^T W\,dW + T.$$

On Episode 1's path, $W_1^2 = 0.444$, the chain-rule sum was -0.618, and the gap 1.062 was the sum of squared steps. Itô replaces it by $T = 1$; the remaining 0.062 is the difference between that path's squared steps and T, which vanishes as the steps shrink.

The log price

For a share with $dS = \mu S\,dt + \sigma S\,dW$, take $f(x) = \ln x$: $f' = 1/x$, $f'' = -1/x^2$, and $(dS)^2 = \sigma^2 S^2\,dt$. So

$$d\ln S = \frac{dS}{S} - \frac{1}{2}\frac{\sigma^2 S^2}{S^2}\,dt = \left(\mu - \tfrac{1}{2}\sigma^2\right)dt + \sigma\,dW.$$

With $\mu = 8\%$ and $\sigma = 20\%$ the log drift is 6.0%, a quarter less than μ. Simulating S step by step from dS itself, with no formula, 100,000 paths give an average log return of 6.0%. The Itô term is of order dt, the same order as the drift: it is never negligible over any horizon.

History

Kiyosi Itô published the formula in 1951 (*On a formula concerning stochastic differentials*, Nagoya Mathematical Journal 3). Wolfgang Doeblin had written a version into the sealed envelope he deposited with the Académie des sciences in 1940, opened in 2000 (Part 1, Chapter 1); hence the name Itô-Doeblin.

Bip's rule. Taylor to second order, and $(dW)^2 = dt$.

Try this. What is $d(W^3)$?

Sources

Companion notebook: `s5e15_ito_doeblin.ipynb`. Every number in this chapter is computed by it.

1. Université Paris-Saclay (author's page), *J.-M. Bismut, address at the homage to Wolfgang (Vincent) Doeblin: the pli cacheté deposited 26 February 1940, opened 18 May 2000*. https://imo.universite-paris-saclay.fr/~jean-michel.bismut/media/files/BBAWAdS%20Bismut.pdf
2. Cambridge University Press (Crossref record), *On a formula concerning stochastic differentials (K. Itô), Nagoya Mathematical Journal 3 (1951), 55-65*. https://doi.org/10.1017/S0027763000012216

CHAPTER 8 · VIDEO EPISODE 16

Two Brownian Motions

Correlated Brownian motions multiply to rho dt, which adds a term to the product rule and to the Itô-Doeblin formula in several dimensions. Lévy's characterisation lets us recover the independent driver.

Last chapter's question

The answer is $d(W^3) = 3W^2\,dW + 3W\,dt$: $f' = 3x^2$ and $\frac{1}{2}f'' = 3x$. So $W_t^3 - 3\int_0^t W\,du$ is a martingale, as Chapter 4 found.

Building correlation

Let W_1 and B be independent Brownian motions and $\rho \in [-1, 1]$. Then

$$W_2 = \rho\, W_1 + \sqrt{1-\rho^2}\; B$$

is a Brownian motion: it starts at 0, has independent normal increments with variance $(\rho^2 + 1 - \rho^2)\Delta t = \Delta t$, and continuous paths. Its correlation with W_1 is $\mathrm{Cov}(W_1(t), W_2(t))/t = \rho$. This is the two-dimensional case of the **Cholesky** factorisation $C = LL^\top$ of a correlation matrix: multiply a vector of independent Brownian motions by L.

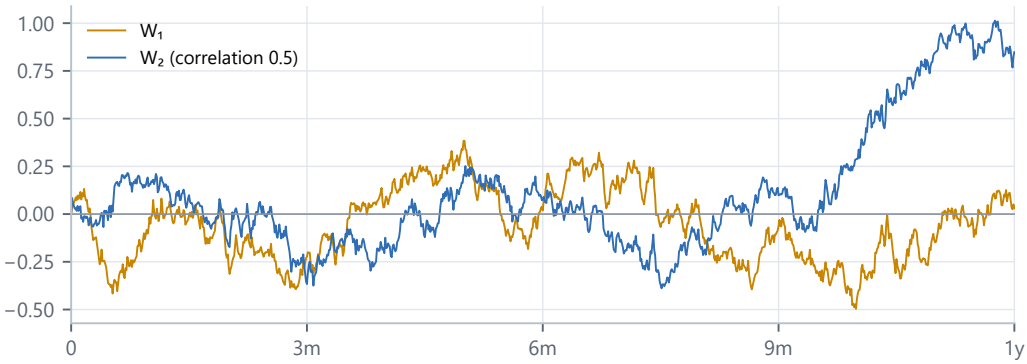


Figure 8.1. One simulated pair with ρ = 0.5, 2^16 steps over a year.

Cross variation

The **cross variation** of two processes is the limit of $\sum \Delta X_i \, \Delta Y_i$. For our pair,

$$\sum \Delta W_1 \, \Delta W_2 = \rho \sum (\Delta W_1)^2 + \sqrt{1-\rho^2} \sum \Delta W_1 \, \Delta B \;\to\; \rho \, t + 0,$$

because the second sum has mean zero and variance $\sum (\Delta t)^2 \to 0$. On the pair above:

The cross variation settles on ρt = 0.5.

STEPS	Σ ΔW₁ ΔW₂
16	0.202
256	0.390
4,096	0.516
65,536	0.503

The multiplication table gains a row: $dW_1 \cdot dW_2 = \rho \, dt$ (and $dW_1 \cdot dB = 0$ for independent motions).

The product rule

For any two paths and any partition, exactly:

$$a_T b_T - a_0 b_0 = \sum a_{t_i} \, \Delta b_i + \sum b_{t_i} \, \Delta a_i + \sum \Delta a_i \, \Delta b_i,$$

since $a_{i+1} b_{i+1} - a_i b_i = a_i \Delta b_i + b_i \Delta a_i + \Delta a_i \Delta b_i$ (the notebook checks it to 1.1e-16). In the limit the left sums become Itô integrals and the last sum becomes the cross variation:

$$d(W_1 W_2) = W_1 \, dW_2 + W_2 \, dW_1 + \rho \, dt.$$

Taking expectations, the Itô integrals vanish and $E[W_1(t) W_2(t)] = \rho t$: 0.497 over 20,000 simulated pairs at $t = 1$. More generally, for $f(X^1, \dots, X^d)$ of Itô processes,

$$df = \sum_i f_i \, dX^i + \tfrac{1}{2} \sum_{i,j} f_{ij} \, dX^i \, dX^j,$$

where $dX^i \, dX^j = \sigma_i \sigma_j \rho_{ij} \, dt$.

Lévy's characterisation

Lévy's characterisation. A continuous martingale M with $M_0 = 0$ and quadratic variation $[M]_t = t$ is a Brownian motion. It lets us undo the mix: $B = (W_2 - \rho W_1)/\sqrt{1-\rho^2}$ is a continuous martingale with $[B]_t = (1 - 2\rho^2 + \rho^2)\, t/(1-\rho^2) = t$,

so it is a Brownian motion, and its cross variation with W_1 is $(\rho - \rho)\, t/\sqrt{1-\rho^2} = 0$. On the simulated pair, $[B]_1 = 0.993$ and the cross variation with W_1 is 0.004; over 20,000 pairs, the correlation of B_1 with $W_1(1)$ is -0.009.

Where it shows up

The $\rho\, dt$ term sits inside every multi-asset model: spread and basket options, quantos (a share paid in another currency), and stochastic volatility, where a price and its own variance are correlated (the Heston model, *Options from Zero*, Part 3, Chapter 8).

Part 2 in one paragraph

Brownian motion is pinned down by four properties and is the limit of random walks. Its paths have no slope and infinite length, but their squared steps add up to time, which makes volatility measurable and drift not. Curved functions of it need a correction to become martingales. The Itô integral samples at the left end of each step, has mean zero and obeys the isometry, and the Itô-Doeblin formula fixes the chain rule with half the second derivative times dt. Part 3 uses all of it for stochastic differential equations, Girsanov's theorem and pricing.

Bip's rule. Multiply the noises and you get $\rho\, dt$.

Try this. The Part 2 checkpoint, at the back of the book.

Sources

Companion notebook: `s5e16_two_brownian_motions.ipynb`. Every number in this chapter is computed by it.

1. Wikipedia, *Wiener process: Lévy characterization (a continuous martingale with W_0 = 0 and quadratic variation t is a Wiener process)*. https://en.wikipedia.org/wiki/Wiener_process

Exercises with answers

Five exercises per chapter: three by hand and two in code. They are the same exercises as at the end of each chapter's notebook, where the code solutions run and check themselves; try each before reading its answer. "Section" in an exercise means a section of that chapter's notebook.

Chapter 1: Brownian Motion, Defined

Exercise 1.1 *(by hand)*. What is $\mathrm{Var}(W_3 - W_1)$?

Answer. The increment over $[1, 3]$ is normal with variance $3 - 1 = 2$.

Exercise 1.2 *(by hand)*. What is $\mathrm{Cov}(W_1, W_4 - W_1)$?

Answer. 0: the increment after time 1 is independent of W_1.

Exercise 1.3 *(by hand)*. Show that $\mathrm{Cov}(W_s, W_t) = s$ for $s < t$ using only properties 2 and 3.

Answer. $\mathrm{Cov}(W_s, W_t) = \mathrm{Cov}(W_s, W_s) + \mathrm{Cov}(W_s, W_t - W_s) = \mathrm{Var}(W_s) + 0 = s$.

Exercise 1.4 *(in code)*. Plot the Kolmogorov distance of Section 4 against the number of steps on log-log axes. What is the slope?

Solution 1.4, from the notebook

```
x = np.log([r["steps"] for r in ks]); y = np.log([r["gap"] for r in ks])
slope = np.polyfit(x, y, 1)[0]
assert -0.6 < slope < -0.4
print(f"slope {slope:.3f}: the gap shrinks like 1/sqrt(n)")
```

What it prints. `slope -0.497: the gap shrinks like 1/sqrt(n)`

Exercise 1.5 *(in code)*. Check the scaling property for $c = 1/2$: what is the variance of $W_{0.5}/\sqrt{0.5}$?

Solution 1.5, from the notebook

```
v = float((Wc[:, 50] / math.sqrt(0.5)).var())
assert abs(v - 1) < 0.04
print(f"variance of W_0.5 / sqrt(0.5): {v:.3f}")
```

What it prints. `variance of W_0.5 / sqrt(0.5): 0.997`

Chapter 2: Brownian Paths Are Strange

Exercise 2.1 *(by hand)*. For a normal Z with mean 0 and variance h, show $E|Z| = \sqrt{2h/\pi}$.

Answer. $E|Z| = 2\int_0^\infty z \frac{1}{\sqrt{2\pi h}} e^{-z^2/2h}\, dz = \frac{2}{\sqrt{2\pi h}} \cdot h = \sqrt{2h/\pi}$.

Exercise 2.2 *(by hand)*. If a path had finite length L, show that its sum of squared steps $\sum(\Delta W)^2$ would go to 0 as the steps shrink. Why does this rule Brownian motion out?

Answer. $\sum(\Delta W)^2 \le \max|\Delta W| \cdot \sum|\Delta W| \le \max|\Delta W| \cdot L \to 0$, since a continuous path has $\max|\Delta W| \to 0$. Brownian squared steps add up to T (Episode 1), so its length can't be finite.

Exercise 2.3 *(by hand)*. To make the first $1/10{,}000$ of the year look like a whole-year Brownian path, by how much must you stretch time, and height?

Answer. Time by $10{,}000$ and height by $\sqrt{10{,}000} = 100$.

Exercise 2.4 *(in code)*. Fit a line to log(average slope) against log(h) from Section 3. What slope do you expect?

Solution 2.4, from the notebook

```
x = np.log([r["h"] for r in slopes]); y = np.log([r["slope"] for r in slopes])
b = np.polyfit(x, y, 1)[0]
assert -0.55 < b < -0.45
print(f"slope of the fit {b:.3f}: the typical slope grows like h^(-1/2)")
```

What it prints. `slope of the fit -0.497: the typical slope grows like h^(-1/2)`

Exercise 2.5 *(in code)*. Count how often the fine path changes sign in the first 1% of the year, and in the whole year. What does that say about the zeros of a Brownian path?

Solution 2.5, from the notebook

```
s = np.sign(wf[1:])                       # skip W_0 = 0, which has no sign
first = int((np.diff(s[: FINE // 100]) != 0).sum())
whole = int((np.diff(s) != 0).sum())
assert (first, whole) == (zoom["zeros"]["hundredth"], zoom["zeros"]["whole"])
print(f"sign changes: {first} in the first 1% of the year (an even spread would give about
{whole // 100}), {whole} in the whole year")
```

What it prints. `sign changes: 27 in the first 1% of the year (an even spread would give about 10), 1093 in the whole year`

Chapter 3: Quadratic Variation: What You Can Measure

Exercise 3.1 *(by hand)*. What is the quadratic variation of $3W$ over $[0, 2]$? Of $W_t + 5t$ over $[0, 1]$?

Answer. $9 \times 2 = 18$; and 1: the drift adds nothing.

Exercise 3.2 *(by hand)*. Why does $\sum |\Delta W\, \Delta t|$ go to zero?

Answer. $\sum |\Delta W|\, \Delta t \le \max |\Delta W| \cdot \sum \Delta t = \max |\Delta W| \cdot T$, and the largest step of a continuous path goes to 0.

Exercise 3.3 *(by hand)*. With $\sigma = 20\%$, how many years of data give a drift estimate with standard deviation 2%?

Answer. $\sigma/\sqrt{T} = 2\%$ needs $\sqrt{T} = 10$: a hundred years.

Exercise 3.4 *(in code)*. Compare the realised volatility with $\mu = 0$ and $\mu = 50\%$ on the same random draws, sampled daily and every 15 minutes. Why does the gap shrink?

Solution 3.4, from the notebook

```
gaps = {}
for n in (252, 252 * 26):        # daily, every 15 minutes
    z = rng.standard_normal((1_000, n))
    v0 = realised_vol((0 - 0.5 * sig ** 2) / n + sig * math.sqrt(1 / n) * z, 1.0)
    v5 = realised_vol((0.5 - 0.5 * sig ** 2) / n + sig * math.sqrt(1 / n) * z, 1.0)
    gaps[n] = float(v5.mean() - v0.mean())
assert gaps[252 * 26] < gaps[252] / 10       # the drift's squared steps are (mu dt)^2: they
vanish as dt shrinks
print(f"drift adds {gaps[252]:.4f} to realised vol sampled daily, {gaps[252 * 26]:.5f} every
15 minutes")
```

What it prints. `drift adds 0.0023 to realised vol sampled daily, 0.00009 every 15 minutes`

Exercise 3.5 *(in code)*. Estimate the quadratic variation of W_t^2 on the fine path. Compare it with $\int_0^1 4W_t^2\, dt$ (Episode 15 explains why they match).

Solution 3.5, from the notebook

```
qv_sq = float((np.diff(wf ** 2) ** 2).sum())
integral = float((4 * wf[:-1] ** 2).sum() / FINE)
assert abs(qv_sq / integral - 1) < 0.02
print(f"QV of W^2: {qv_sq:.4f}; integral of 4 W^2 dt: {integral:.4f}")
```

What it prints. `QV of W^2: 0.3279; integral of 4 W^2 dt: 0.3274`

Chapter 4: Martingales from Brownian Motion

Exercise 4.1 *(by hand).* Derive $E[W_t^2 \mid \mathcal{F}_s]$ for $s < t$.

Answer. $W_t^2 = (W_s + D)^2 = W_s^2 + 2W_sD + D^2$. Given $\mathcal{F}_s$, W_s is known, $E[D] = 0$ and $E[D^2] = t - s$: the answer is $W_s^2 + t - s$.

Exercise 4.2 *(by hand).* Use the normal moment generating function $E[e^{aZ}] = e^{a^2v/2}$ for $Z \sim N(0, v)$ to find $E[e^{\sigma W_t}]$.

Answer. $e^{\sigma^2 t/2}$: here $a = \sigma$ and $v = t$.

Exercise 4.3 *(by hand).* Is $W_t + t$ a martingale? Is it a sub- or supermartingale?

Answer. No: $E[W_t + t \mid \mathcal{F}_s] = W_s + t > W_s + s$. It drifts up, so it is a submartingale.

Exercise 4.4 *(in code).* Check by simulation that $W_t^3 - 3tW_t$ averages zero at $t = 0.5, 1, 2$. (It is a martingale too.)

Solution 4.4, from the notebook

```
for t_, i in ((0.5, 50), (1.0, 100), (2.0, 200)):
    v = W[:, i] ** 3 - 3 * t_ * W[:, i]
    assert abs(v.mean()) < 4 * v.std() / math.sqrt(len(v))
    print(f"t = {t_}: mean of W^3 - 3tW = {v.mean():+.4f} (standard error {v.std() /
math.sqrt(len(v)):.4f})")
```

What it prints. `t = 0.5: mean of W^3 - 3tW = -0.0014 (standard error 0.0027)`; `t = 1.0: mean of W^3 - 3tW = -0.0068 (standard error 0.0078)`; `t = 2.0: mean of W^3 - 3tW = -0.0232 (standard error 0.0220)`

Exercise 4.5 *(in code).* Repeat Section 4's grouping test for the exponential martingale $\exp(\sigma W_t - \frac{1}{2}\sigma^2 t)$.

Solution 4.5, from the notebook

```
e = lambda w, t_: np.exp(sig * w - 0.5 * sig ** 2 * t_)
for g in range(1, len(edges)):
    m = grp == g
    later, now = e(W[m, 100], 1.0), e(W[m, 50], 0.5)
    assert abs(later.mean() - now.mean()) < 3 * later.std() / math.sqrt(m.sum())
print("exponential martingale: later average matches today's value in every group")
```

What it prints. `exponential martingale: later average matches today's value in every group`

Chapter 5: Building the Itô Integral

Exercise 5.1 *(by hand)*. Show that the middle sum $\sum \frac{1}{2}(W_i + W_{i+1})(W_{i+1} - W_i)$ equals $\frac{1}{2}(W_T^2 - W_0^2)$ exactly.

Answer. Each term is $\frac{1}{2}(W_{i+1}^2 - W_i^2)$, a difference of squares, and the sum telescopes.

Exercise 5.2 *(by hand)*. Show that the right sum minus the left sum is $\sum(\Delta W_i)^2$.

Answer. Term by term: $W_{i+1}\Delta W_i - W_i\Delta W_i = (W_{i+1} - W_i)\Delta W_i = (\Delta W_i)^2$.

Exercise 5.3 *(by hand)*. What is $\int_0^T 1\, dW$?

Answer. $W_T - W_0 = W_T$: every sampling rule gives the same telescoping sum.

Exercise 5.4 *(in code)*. On the fine path, compare left sums for $\int_0^1 W^2\, dW$ with $\frac{1}{3}W_1^3 - \int_0^1 W\, dt$ (Episode 15 shows why).

Solution 5.4, from the notebook

```
for n in (256, 4_096, 2 ** 20):
    p = wf[:: 2 ** 20 // n]
    lhs = float((p[:-1] ** 2 * np.diff(p)).sum())
    rhs = p[-1] ** 3 / 3 - float(p[:-1].sum() / n)
    print(f"{n:>8} steps: left sum {lhs:+.4f}, formula {rhs:+.4f}")
assert abs(lhs - rhs) < 0.01
```

What it prints. `256 steps: left sum -0.1885, formula -0.1761`; `4096 steps: left sum -0.1753, formula -0.1772`; `1048576 steps: left sum -0.1775, formula -0.1773`

Exercise 5.5 *(in code)*. On 5 new random paths of 2^16 steps, check that the left sum is close to $\frac{1}{2}W_1^2 - \frac{1}{2}$.

Solution 5.5, from the notebook

```
_, P, dP = brownian_paths(5, 2 ** 16, 1.0, np.random.default_rng(seed + 13))
for p, d in zip(P, dP):
    gap = (p[:-1] * d).sum() - (0.5 * p[-1] ** 2 - 0.5)
    assert abs(gap) < 0.03
print("left sums match 1/2 W^2 - 1/2 T on all 5 paths (to within 0.03)")
```

What it prints. `left sums match 1/2 W^2 - 1/2 T on all 5 paths (to within 0.03)`

Chapter 6: No Peeking, No Free Lunch

Exercise 6.1 *(by hand)*. Two steps of length Δt: $H_0 = 1$, $H_1 = W_{\Delta t}$. Compute $E[(H_0\Delta W_0 + H_1\Delta W_1)^2]$ directly and check it equals $E[H_0^2 + H_1^2]\,\Delta t$.

Answer. Cross term $2E[\Delta W_0 \cdot W_{\Delta t}\Delta W_1] = 0$ since ΔW_1 is independent with mean 0. Diagonal: $\Delta t + E[W_{\Delta t}^2]\Delta t = \Delta t + \Delta t^2$, which is $E[H_0^2 + H_1^2]\Delta t$. ✓

Exercise 6.2 *(by hand).* Use the isometry to find $\mathrm{Var}\big(\int_0^T W\,dW\big)$, and check it against Episode 13's formula $\frac{1}{2}W_T^2 - \frac{1}{2}T$.

Answer. Isometry: $\int_0^T t\,dt = T^2/2$. Directly: $\mathrm{Var}(\frac{1}{2}W_T^2) = \frac{1}{4}\mathrm{Var}(W_T^2) = \frac{1}{4}\cdot 2T^2 = T^2/2$. ✓

Exercise 6.3 *(by hand).* Why does the peeking gain grow like $\sqrt{N}$ with the number of trades?

Answer. Each step earns $|\Delta W| \approx \sqrt{2\Delta t/\pi}$, and there are $N = T/\Delta t$ steps: $N\sqrt{2T/(\pi N)} = \sqrt{2TN/\pi}$.

Exercise 6.4 *(in code).* Check the isometry for $H = W^2$: $E[(\int_0^1 W^2 dW)^2] = \int_0^1 E[W_t^4]\,dt = \int_0^1 3t^2\,dt = 1$.

Solution 6.4, from the notebook

```
J = (W[:, :-1] ** 2 * dW).sum(axis=1)
se = float((J ** 2).std() / math.sqrt(P))
assert abs((J ** 2).mean() - 1) < 4 * se
print(f"E[(integral of W^2 dW)^2] = {(J ** 2).mean():.3f} (theory 1, standard error
{se:.3f})")
```

What it prints. `E[(integral of W^2 dW)^2] = 0.896 (theory 1, standard error 0.042)`

Exercise 6.5 *(in code).* A strategy that reacts to the previous step, $H_i = \mathrm{sign}(\Delta W_{i-1})$, uses only the past. Check its gains still average zero.

Solution 6.5, from the notebook

```
Hm = np.concatenate([np.zeros((P, 1)), np.sign(dW[:, :-1])], axis=1)
Gm = (Hm * dW).sum(axis=1)
assert abs(Gm.mean()) < 4 * Gm.std() / math.sqrt(P)
print(f"momentum strategy: mean gain {Gm.mean():+.4f} (standard error {Gm.std() /
math.sqrt(P):.4f})")
```

What it prints. `momentum strategy: mean gain -0.0023 (standard error 0.0070)`

Chapter 7: The Itô-Doeblin Formula

Exercise 7.1 *(by hand).* Find $d(W_t^3)$ and $d(W_t^4)$.

Answer. $d(W^3) = 3W^2\,dW + 3W\,dt$; $d(W^4) = 4W^3\,dW + 6W^2\,dt$.

Exercise 7.2 *(by hand)*. Show that $Y_t = e^{\sigma W_t - \frac{1}{2}\sigma^2 t}$ has $dY = \sigma Y\, dW$ (no dt term), so it is a martingale (Episode 12).

Answer. $f(t,x) = e^{\sigma x - \frac{1}{2}\sigma^2 t}$: $f_t = -\frac{1}{2}\sigma^2 f$, $f_x = \sigma f$, $f_{xx} = \sigma^2 f$. So $dY = -\frac{1}{2}\sigma^2 Y\,dt + \sigma Y\,dW + \frac{1}{2}\sigma^2 Y\,dt = \sigma Y\,dW$.

Exercise 7.3 *(by hand)*. Find $d(tW_t)$.

Answer. $f(t,x) = tx$: $f_t = x$, $f_x = t$, $f_{xx} = 0$, so $d(tW) = W\,dt + t\,dW$: no Itô correction, because f is linear in x.

Exercise 7.4 *(in code)*. Check Itô's formula pathwise for $f = \sin$ on the fine path.

Solution 7.4, from the notebook

```
cs = ito_check(np.sin, np.cos, lambda x: -np.sin(x), wf)
assert abs(cs["exact"] - cs["ito"]) < 0.01
print(f"sin: exact change {cs['exact']:+.4f}, Ito {cs['ito']:+.4f}, chain rule
{cs['chain']:+.4f}")
```

What it prints. `sin: exact change -0.1805, Ito -0.1804, chain rule -0.0958`

Exercise 7.5 *(in code)*. Repeat Section 6 with $\sigma = 40\%$. What drift does the log price have now?

Solution 7.5, from the notebook

```
S4 = np.full(50_000, S0)
r4 = np.random.default_rng(seed + 15)
for _ in range(252):
    S4 = S4 * (1 + mu / 252 + 0.4 * math.sqrt(1 / 252) * r4.standard_normal(S4.size))
m4 = float(np.log(S4 / S0).mean())
assert abs(m4 - (mu - 0.5 * 0.4 ** 2)) < 0.01
print(f"mean log return {m4:+.4f}; Ito says mu - sigma^2/2 = {mu - 0.08:+.4f}")
```

What it prints. `mean log return +0.0022; Ito says mu - sigma^2/2 = +0.0000`

Chapter 8: Two Brownian Motions

Exercise 8.1 *(by hand)*. With $\rho = 0$, what is $d(W_1 W_2)$? What is $E[W_1(t)W_2(t)]$?

Answer. $d(W_1W_2) = W_1\,dW_2 + W_2\,dW_1$: no extra term. $E[W_1W_2] = 0$.

Exercise 8.2 *(by hand)*. What is $\mathrm{Var}(W_1(t) + W_2(t))$ when their correlation is ρ?

Answer. $t + t + 2\rho t = 2(1+\rho)t$.

Exercise 8.3 *(by hand)*. What is the quadratic variation of $W_1 + W_2$ over $[0,t]$?

Answer. $(dW_1 + dW_2)^2 = dt + dt + 2\rho\, dt$, so $2(1+\rho)t$: the same as the variance.

Exercise 8.4 *(in code)*. Build a pair with $\rho = -0.8$ and check the cross variation on a fine grid.

Solution 8.4, from the notebook

```
_, P1, P2 = correlated_paths(1, 2 ** 16, -0.8, np.random.default_rng(seed + 16))
cv = float((np.diff(P1[0]) * np.diff(P2[0])).sum())
assert abs(cv + 0.8) < 0.03
print(f"cross variation {cv:+.3f} (rho = -0.8)")
```

What it prints. `cross variation -0.797 (rho = -0.8)`

Exercise 8.5 *(in code)*. Build three Brownian motions with pairwise correlations 0.5, 0.3 and 0.2 from a 3 × 3 Cholesky factor, and check their sample correlations.

Solution 8.5, from the notebook

```
C = np.array([[1, 0.5, 0.3], [0.5, 1, 0.2], [0.3, 0.2, 1]])
L = np.linalg.cholesky(C)
Z = np.random.default_rng(seed + 160).standard_normal((3, 200_000))
X = L @ Z
est = np.corrcoef(X)
assert np.abs(est - C).max() < 0.01
print(np.round(est, 3))
```

What it prints. `[[1. 0.502 0.3 ]` `[0.502 1. 0.203]` `[0.3 0.203 1. ]]`

Checkpoint, glossary and further reading

Part 2 checkpoint

Try these before Part 3. The answers follow.

1. What is $\mathrm{Cov}(W_2, W_7)$?
2. What is the quadratic variation of $2W_t + 5t$ over $[0, 3]$?
3. Find $d(W^4)$, and the martingale it implies.
4. What is $\mathrm{Var}\big(\int_0^1 W_t^2\, dW_t\big)$?

Answers.

1. $\min(2, 7) = 2$ (Chapter 1).
2. $2^2 \times 3 = 12$: the drift adds nothing (Chapter 3).
3. $d(W^4) = 4W^3\, dW + 6W^2\, dt$ (Chapter 7), so $W_t^4 - 6\int_0^t W_u^2\, du$ is a martingale (Chapter 4). Simulated over 20,000 paths at $t = 1$ it averages -0.040, within one standard error (0.052) of zero.
4. By the isometry, $\int_0^1 E[W_t^4]\, dt = \int_0^1 3t^2\, dt = 1$ (Chapter 6); simulated: 0.956.

Glossary

Adapted. A process whose value at time t is known at time t.

Brownian motion (Wiener process). Starts at 0; independent, normal increments with variance equal to the elapsed time; continuous paths.

Cholesky factorisation. Writing a correlation matrix as $LL^\top$; multiplying independent Brownian motions by L gives correlated ones.

Cross variation. The limit of $\sum \Delta X\, \Delta Y$; for correlated Brownian motions, ρt.

Donsker's theorem. Scaled random walks converge to Brownian motion as whole paths: the functional central limit theorem.

Exponential martingale. $\exp(\sigma W_t - \frac{1}{2}\sigma^2 t)$.

Itô integral. The mean-square limit of left-point sums $\sum H_{t_i}\,\Delta W_i$ for an adapted H.

Itô isometry. $E[(\int H\,dW)^2] = E[\int H^2\,dt]$.

Itô process. $dX = \mu\,dt + \sigma\,dW$ with adapted μ and σ.

Itô-Doeblin formula. $df(t,X) = f_t\,dt + f_x\,dX + \frac{1}{2}f_{xx}\,(dX)^2$, with $(dX)^2 = \sigma^2\,dt$.

Jensen's inequality. For convex f, $E[f(X)] \geq f(E[X])$.

Lévy's characterisation. A continuous martingale from 0 with quadratic variation t is a Brownian motion.

Multiplication table. $dW \cdot dW = dt$, $dW \cdot dt = dt \cdot dt = 0$, $dW_1 \cdot dW_2 = \rho\,dt$.

Quadratic variation. The limit of $\sum(\Delta X)^2$; $[W]_t = t$, $[\sigma W]_t = \sigma^2 t$.

Realised volatility. The square root of the sum of squared log returns over a period, annualised.

Self-similarity. A Brownian path, zoomed in by c in time and $\sqrt{c}$ in height, is again a Brownian path.

Simple integrand. A process constant on each step of a partition, known at the start of the step.

Stratonovich integral. The limit of middle-point sums; keeps the school chain rule but is not a martingale.

Total variation (length). The limit of $\sum |\Delta X|$; infinite for Brownian motion.

Further reading

This book is written from its own notebooks; these are excellent companions. None of their examples or exercises are used here.

- Steven E. Shreve, *Stochastic Calculus for Finance II: Continuous-Time Models*: Chapter 3 for Chapters 1 to 4, Chapter 4 for Chapters 5 to 8.
- Bernt Øksendal, *Stochastic Differential Equations*: Chapters 2 to 4.
- Fima C. Klebaner, *Introduction to Stochastic Calculus with Applications*: Chapters 3 and 4, a bridge between this book and the rigorous texts.
- Ioannis Karatzas and Steven E. Shreve, *Brownian Motion and Stochastic Calculus*: the full measure-theoretic treatment, for later.

Check the chapter pointers against the edition you have.